SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

QUESTION: 1 **Total Marks:100**

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

Consider a delivery robot operating in a warehouse represented as a 5×5 grid. The robot starts at cell (1,1) and must deliver a package to cell (5,5). Certain cells are blocked due to shelving units: (2,2), (2,3), (3,3), (4,2), and (4,4). The robot can move Up, Down, Left, or Right (no diagonal moves), and each move has a uniform cost of 1.

Answer the following:

a. Model this scenario as a formal intelligent agent problem. Clearly define the agent's PEAS description (Performance measure, Environment, Actuators, Sensors) for this warehouse robot.

b. Classify the type of agent (simple reflex, model-based, goal-based, or utility-based) best suited for this task and justify your choice with reasoning specific to this scenario.

c. Formally define this as a search problem by specifying: Initial State, Goal State, Actions, Transition Model, and Path Cost function.

d. Apply Breadth-First Search (BFS) to find a path from (1,1) to (5,5). Show the step-by-step expansion of nodes (the frontier/queue at each step) until the goal is reached, and state the final path along with its total cost.

e. Apply Depth-First Search (DFS) to the same problem. Show the step-by-step expansion of nodes (the stack at each step) until the goal is reached, and state the final path along with its total cost.

f. Compare BFS and DFS for this specific scenario in terms of completeness, optimality, time complexity, and space complexity, and justify which algorithm is more suitable for this warehouse delivery problem.

Marks Distribution Summary:

Part	Concept Assessed	Marks
a	PEAS / Agent Modeling	15
b	Agent Type Classification	15
c	Search Problem Formulation	15
d	BFS Application	25
e	DFS Application	20
f	Comparative Analysis	10
Total		**100**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem